

DIVYARTH JAIN

Electronics & PCB Design · Embedded Systems · Robotics

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EDUCATION

VIT Bhopal University, B.Tech Electronics & Communication Engineering (AI & Cybernetics) Sept 2023 – present
Model Higher Secondary School, Senior Secondary (PCM) 2023

PUBLICATION

A. Chouksey, D. Joshi, **D. Jain**, A. Beohar, A. Rawat, K. Khare, “Stability-Aware NC-CMOS Logic Design Through Ferroelectric Capacitance Matching: A Landau–Khalatnikov-Based Parametric Study.” *Accepted, to appear* (Paper ID 2381).
• One sizing rule for hysteresis-free negative-capacitance logic, $r = C_{\text{MOS,max}}/|C_{\text{FE}}| < 1$, verified in ngspice (PTM 45 nm BSIM4, Landau–Khalatnikov ferroelectric): it matched 267/270 PVT-corner and 1952/2000 Monte-Carlo runs.

PCB & ELECTRONICS

Six-Servo Driver and Sensor-Feedback Board Sept 2026

KiCad 10, PCA9685, MCP6004, MCP3008, Raspberry Pi

- Schematic for a robot-arm board: 6-channel PWM drive over I²C, plus 12 feedback channels (joint position and motor current) over SPI.
- Current sense: 0.15 Ω shunt into a non-inverting stage, $G = 1 + 19.1 \text{ k}/1 \text{ k} = 20.1$, so 1 A maps to 3.02 V inside the 3.3 V ADC range; 1 k/100 nF filter at $f_c \approx 1.6 \text{ kHz}$.
- Power input: 5 A polyfuse, IRF4905 P-FET reverse-polarity protection, SMBJ5.0A TVS and 1000 μF bulk, budgeted for 1 A per servo. Schematic at Rev 3; layout next.

Prometheus S3 board (EasyEDA, 4-layer, JLCPCB rules): ESP32-S3 with two USB-C ports (native USB + CP2102N UART), TVS ESD protection, auto-programmer flashing. **Also:** ESP32 ToF distance meter (VL53L0X, OLED, RGB status) · AC energy meter with surge auto-cutoff · Verilog motor controller on FPGA vs. MCU-interrupt control.

ROBOTICS

Capstone (DSN4091): Physics-Aware Robot Brain 2026 – present

ST3215-HS bus servos, Raspberry Pi 4B, Pi Cam 2, PyTorch, Isaac Sim, MuJoCo

- A small world model on the robot that predicts its next state and retrains itself on board when its own prediction error (“surprise”) shows the world changed.
- Simulation, 10 seeds: after a sudden change, surprise drops to 42 vs. 114 for a frozen model, using about 4× less training compute than always-on learning (376 vs. 1570 gradient steps).

ArcheoHex Hexapod 2024

18 servos, Raspberry Pi, Python, inverse kinematics, MobileNet

- Built an 18-servo hexapod for tight, uneven cavities; wrote the inverse-kinematics engine from scratch for gait adaptation on rough ground.
- On-board MobileNet object detection at 88.7% accuracy in low light.

Hexapod v2 2nd–3rd year

RP2040 Zero, 2× PCA9685, ESP32, 3 UBECs, 3D-printed frame

- Ported open-source hexapod firmware to my own controller stack and added smoother gaits; ESP32 runs the HUD and an Xbox controller link, three UBECs keep the servo rail stable.

Eco-SLAM Rover

ROS2 Humble, Nav2, ESP32, Raspberry Pi 4, 8× VL53L0X

- 360° ToF ring instead of LiDAR (about 80% lower sensor cost); custom ROS2 driver publishes the 8 streams as one LaserScan for Nav2.

EXPERIENCE & LEADERSHIP

QSC Solutions, AI Product Intern May – Jul 2025

Video-avatar personalization: tuned the personalization logic and lip-sync pipeline; findings fed the feature roadmap.

A.I.E.M. Club, VIT Bhopal, President Jul 2024 – present

Workshops on PCB design, embedded AI and robotics for 100+ students; teaches juniors ESP32-S3/C3 in MicroPython. Student volunteer at IEEE IECS 2026 (posters, banners, session links).

SKILLS

PCB & EDA	KiCad 10 (schematic, layout, libraries), LTspice, ngspice (BSIM4), Cadence/Synopsys with SCL 180 nm PDK (lab setup)
Embedded	ESP32 (ESP-IDF, Arduino, MicroPython), STM32 HAL, FreeRTOS, Raspberry Pi; I ² C, SPI, I ² S, UART, CAN
HDL & Code	Verilog, VHDL, C, C++, Python, MATLAB, Bash
Robotics & ML	ROS2, Nav2, Isaac Sim, MuJoCo, PyTorch, inverse kinematics

ACHIEVEMENTS

Represented India at an international sailing championship; 3× national sailing, 1× national snooker · Caterpillar Autonomy Challenge finalist (IIT Madras, rover team) · MathWorks DSP certificates (filter design, spectral analysis, time-frequency analysis)